HR ANALYTICS DASHBOARD

Project Report

Student Name	Anurag Singh Chauhan
Registration Number	PIET24CD008
Project Title	HR Analytics Dashboard using Power BI

This report describes an interactive dashboard designed to summarize employee headcount, attrition, age distribution, department-level exits, education field, and job satisfaction. The screenshots supplied with the project are included as evidence of the dashboard views.

Prepared from the dashboard screenshots provided

1. Project Overview

Human Resources (HR) analytics uses employee data to help an organization understand its workforce and make better people-related decisions. This project presents HR information in a Power BI-style dashboard so that users can quickly compare overall workforce figures and explore patterns in employee attrition.

2. Dataset Description

The dashboard appears to use an employee-level HR attrition dataset. Each row is expected to represent an employee, with fields covering employee status and selected demographic and job details. The screenshots show metrics for 1,470 employees in the overall view, including 237 attrition cases and 1,233 active employees.

Data area	Examples of fields represented in the dashboard
Employee status	Attrition status; active/left employee count
Demographics	Age, gender
Work and organization	Department, job role
Education	Education field and education level
Employee experience	Job satisfaction rating

Note: The original source file and full column list were not provided. Dataset details above are described from the visible dashboard; the exact source, row-level schema, and cleaning log should be verified against the Power BI file before formal submission.

3. Project Objectives

• Show key workforce indicators: overall employees, attrition, attrition rate, active employees, and average age.

• Compare attrition across departments and education fields.

• Explore the distribution of employees by age band and gender.

• Present job satisfaction ratings by job role in a compact table.

• Allow users to filter results by education level and interact with visuals to investigate specific groups.

4. Data Cleaning and Preparation

The screenshots do not expose the transformation steps, so the following is a concise, recommended cleaning checklist for this dashboard rather than a claim that each step was confirmed in the source file:

• Check for duplicate employee records using the employee identifier.

• Review blank or null values in fields used by visuals, such as age, department, gender, education, and attrition.

• Standardize category labels and remove accidental spaces or inconsistent spellings.

• Confirm data types: age and ratings as numeric values; employee categories as text; attrition as a consistent Yes/No value.

• Create consistent age bands and validate that employee counts reconcile with the source data.

• Check that slicers and visual totals respond correctly to filters.

5. Data Model

A simple single-table model is suitable if the dataset contains one record per employee. The employee table can feed the KPI cards, pie chart, stacked column chart, satisfaction matrix, and education-field bar chart. The

education-level slicer filters the same table, causing the displayed totals and charts to update together. If the project uses separate lookup tables, a star schema can be used with an Employees fact table connected to dimensions such as Department, Job Role, Education, and Date (if date fields exist).

Suggested relationship design: Employees (many/one row per employee) → lookup dimensions by stable keys; use single-direction filtering from dimensions to Employees wherever practical.

6. DAX Measures

The following sample measures reproduce the KPI logic visible in the screenshots. They assume a table named **Employees** and a text column named **Attrition** containing “Yes” or “No”. Replace table/column names to match the actual model.

Overall Employees

```
Overall Employees = DISTINCTCOUNT(Employees[EmployeeNumber])
```

Attrition

```
Attrition = CALCULATE([Overall Employees], Employees[Attrition] = "Yes")
```

Active Employees

```
Active Employees = CALCULATE([Overall Employees], Employees[Attrition] = "No")
```

Attrition Rate

```
Attrition Rate = DIVIDE([Attrition], [Overall Employees], 0)
```

Average Age

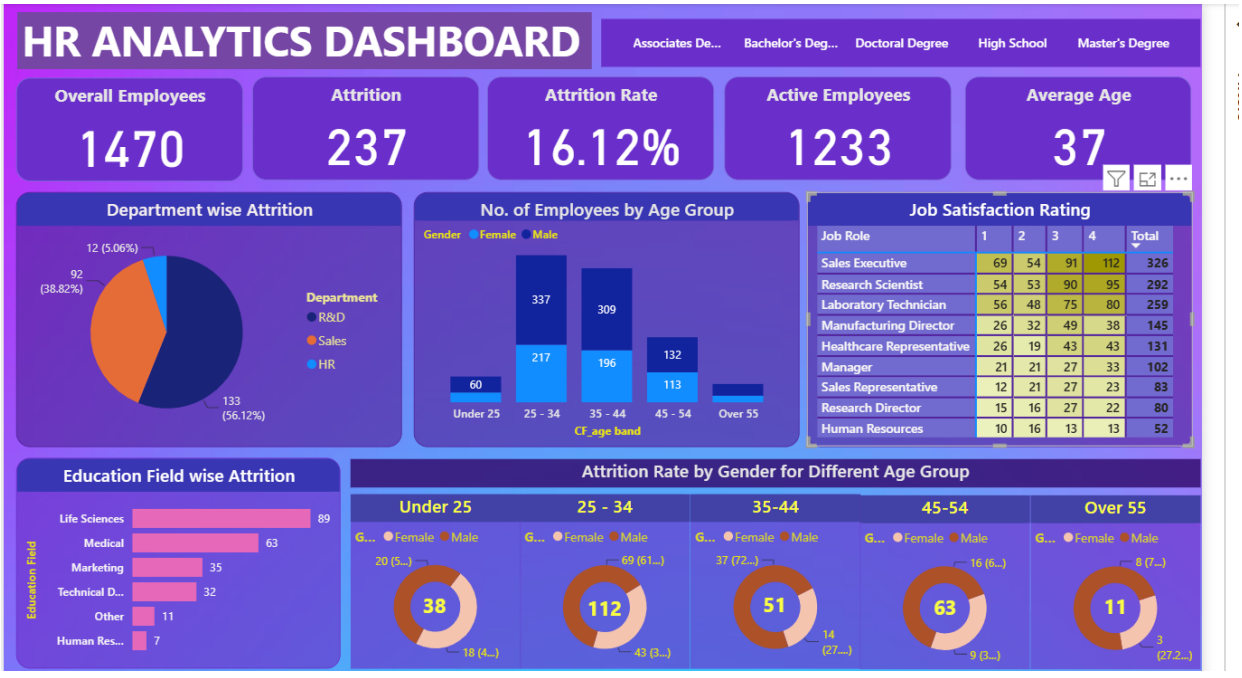
```
Average Age = AVERAGE(Employees[Age])
```

Format Attrition Rate as Percentage with two decimal places. If the source does not contain a unique employee number, use COUNTROWS(Employees) only after verifying that each row represents exactly one employee.

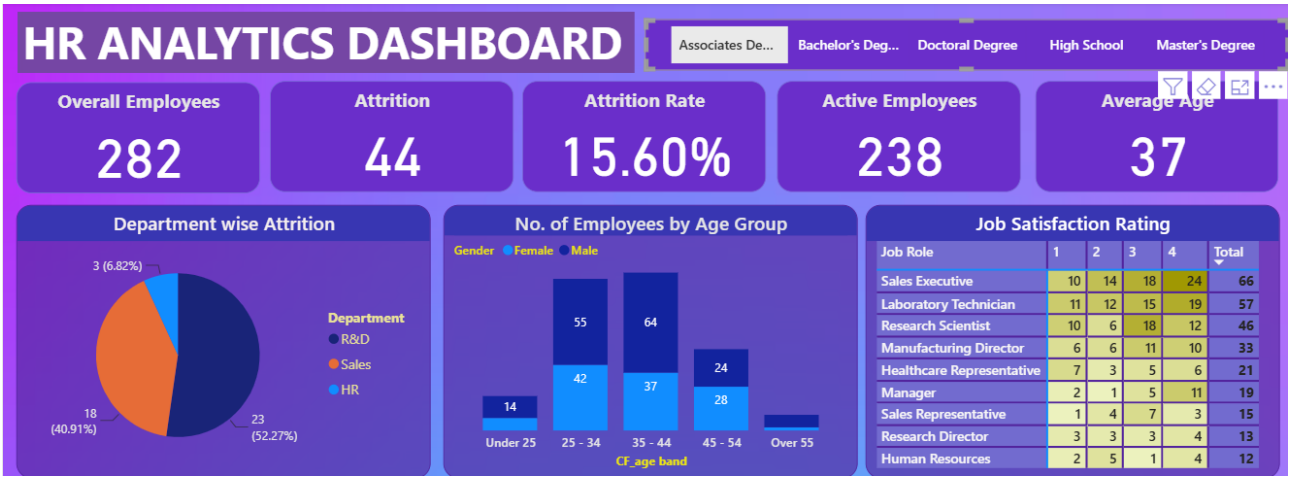
7. Dashboard Screenshots

Screenshot A shows the overall workforce view. Screenshot B shows a filtered view, where the education-level slicer is selected; values therefore differ from the overall view.

Screenshot A — Overall employee view



Screenshot B — Filtered education-level view



Job Role	1	2	3	4	Total
Sales Executive	10	14	18	24	66
Laboratory Technician	11	12	15	19	57
Research Scientist	10	6	18	12	46
Manufacturing Director	6	6	11	10	33
Healthcare Representative	7	3	5	6	21
Manager	2	1	5	11	19
Sales Representative	1	4	7	3	15
Research Director	3	3	3	4	13
Human Resources	2	5	1	4	12

8. Key Findings

• **Overall workforce:** the main dashboard reports 1,470 employees, 237 attrition cases, 1,233 active employees, and an average age of 37.

• **Attrition rate:** the overall rate displayed is 16.12%, meaning roughly 16 out of every 100 employees left in the period represented by the dataset.

• **Department:** R&D has the largest share of attrition in the overall pie chart (133; 56.12%), followed by Sales (92; 38.82%) and HR (12; 5.06%).

• **Age distribution:** the 25–34 and 35–44 age bands have the tallest employee bars, indicating that much of the workforce is in these age groups.

• **Job satisfaction:** Sales Executive has the largest total shown in the satisfaction table (326 in the overall view), followed by Research Scientist (292) and Laboratory Technician (259). These totals are counts across ratings, not average satisfaction scores.

• **Filtered view:** the second screenshot shows 282 employees, 44 attrition cases, a 15.60% attrition rate, 238 active employees, and average age 37 for the selected education category.

9. Conclusion

The HR Analytics Dashboard brings several workforce measures into one interactive report. It helps a user compare attrition by department, age, gender, education, and job role, while the slicer supports focused analysis of employee subgroups. The overall view indicates a 16.12% attrition rate, with R&D contributing the largest number of attrition cases in the displayed department breakdown. HR teams could use these patterns to decide where to investigate retention, workload, career growth, and employee experience in more detail.

Limitation: the dashboard shows descriptive patterns, not the reasons employees leave. Any retention action should be based on further analysis and discussions with employees, and the measures should be validated against the original dataset and Power BI model.

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